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[Chap 7] Model Performance Metrics

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Type I and Type II errors

- The Wolf and the Shepherd Boy –

  The first time, the villagers ran to the shepherd at the sound of “The wolf has appeared!” even though there was no wolf. – The second time, the villagers ran to the shepherd at the sound of “The wolf has appeared!” even though there was no wolf. – The third time, the villagers did not believe the sound of “The wolf has appeared!” even though the wolf had appeared. ÿ The villagers made three judgments, and were wrong all three times.

- Null Hypothesis and Alternative

  Hypothesis – Alternative Hypothesis: A hypothesis that, upon observing the data, there is a significant difference from the existing situation (such as a different mean).

  – Null Hypothesis: A hypothesis that there is no significant difference from the existing situation. – The villagers hearing the story of the shepherd boy and thinking that 'there is a wolf' is the alternative hypothesis, while 'there is no wolf' is the alternative hypo Thinking that becomes the null hypothesis.

- Villagers' Error – First and Second

  Cases: Predicted that there is a wolf (positive) but was wrong (actually negative) ÿ Type I error, false alarm – Third case: Predicted no wolf (negative) but was wrong (actually positive) ÿ Type II error, missed detection

  – In statistics, the p-value is defined as the probability of a Type I error.

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Confusion matrix

• Example of a True/False Quiz: Scoring sheet for 10 questions

〈표 7-1〉 OX퀴즈 채점표

구분	1번	2번	3번	4번	5번	6번	7번	8번	9번	10번
정답	X	X	X	X	O	O	O	X	X	O
답안지	X	X	X	O	O	O	X	X	O	O
정오구분	정답	정답	정답	오답	정답	정답	오답	정답	오답	정답

– This student got 7 out of 10 questions right and missed 3, scoring 70 points – 2 types of missed questions

(cases where the correct answer was O but X was written, and cases where the correct answer was X but O was written),

There are two types of correct answers (cases where the answer is O but written as O, and cases where the answer is X but written as X) –

Scoring sheet by correct/incorrect answer type

〈표 7-2〉 정오답 유형별 채점표

답안지 / 정답	X	O
X	4	2
O	1	3

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Confusion Matrix (continued)

- The correct answer is called the actual value, and the student's answer is called the predicted value, and

  If we represent O(True) as the number 1 and X(False) as the number 0, it is as follows:

〈표 7-3〉 OX퀴즈에 대한 혼동행렬

Actual \ Predicted	0	1
0	4	2
1	1	3

– This is called the confusion matrix

- Confusion Matrix Calculation Code

```
from sklearn.metrics import confusion_matrix
y_true = [0,0,0,0,1,1,1,0,0,1]
y_pred = [0,0,0,1,1,1,0,0,1,1]
print(confusion_matrix(y_true, y_pred))
```

```
[[4 2]
 [1 3]]
```

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TP, TN, FP, FN

- The two correct cases and two incorrect cases are distinguished and referred to as TP, TN, FP, and FN, respectively. –

  The first letters, T or F, stand for True and False, indicating whether the result was actually correct or incorrect. –

  The last letters, P and N, stand for Positive and Negative, respectively, referring to my answer sheet—that is, the value I predicted.

- TP, TN, FP, FN in the confusion matrix below

〈표 7-4〉 혼동행렬의 예시

Actual \ Predicted	0 (Negative)	1 (Positive)
0 (Negative)	4 (TN)	2 (FP)
1 (Positive)	1 (FN)	3 (TP)

– TP (True Positive): The number of times the answer was "Positive" and was correct (TP=3)

– TN(True Negative): Number of times the answer was Negative but correct (TN=4)

– FP(False Positive): The number of times the answer was "Positive" but was incorrect (FP=2)

– FN(False Negative): Number of times the answer is Negative and incorrect (FN=1)

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Accuracy

- Ratio of the number of correct answers to the total number of predicted answers
  - Same meaning as an OX test score
- Accuracy = (TP + TN) / (TP + TN + FP + FN) – Divided by the diagonal values of the confusion matrix by the total sum – In the figure below, (3 + 4)/10 = 70%

Actual \ Predicted	0 (Negative)	1 (Positive)
0 (Negative)	4 (TN)	2 (FP)
1 (Positive)	1 (FN)	3 (TP)

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Problems with accuracy

- High accuracy does not necessarily mean it is a good model

- Example) If Model A, a special pair of glasses designed to identify aliens, has an accuracy of 99.1%, is that considered good performance?
  - The confusion matrix of these special glasses

〈표 7–5〉 외계인 판별 안경 A의 혼동행렬

Actual \ Predicted	0 (지구인)	1 (외계인)
0 (지구인)	990	0
1 (외계인)	9	1

  - Only 1 out of 10 aliens can be correctly identified –

  Despite performance being poor by common sense standards, the accuracy was found to be 99.1%

- This problem occurs when positives are very rare among the total negatives.
  - Since critical judgments often involve identifying rare targets (e.g., virus scanning, 6-sigma level defect rates), relying solely on accuracy is risky.

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Precision and recall

- Precision – The proportion of correct predictions out of all positives predicted by the model – Precision = TP / (TP + FP) – High precision means a low Type I error, i.e., fewer false alarms – In the previous alien detection glasses example, Precision = 100% (1 detection was correct)

- Recall – Measures the percentage of actual positive values detected by the model – Recall = TP / (TP+FN) – High recall means low Type II error, i.e., fewer missed detections – In the previous alien detection glasses example, accuracy = 1 / (1 + 9) = only 10%

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Necessity of comprehensive performance evaluation

• Example) Performance measurement of Alien Detection Glasses Model B

〈표 7-6〉 외계인 판별 안경 B의 혼동행렬

Actual \ Predicted	0 (지구인)	1 (외계인)
0 (지구인)	981	9
1 (외계인)	1	9

– Accuracy = (981 + 9) / 1000 = 99.0% – Precision

= 9 / (9 + 9) = 50% – Recall = 9 / (1 + 9)

= 90%

지표	모델 A	모델 B
정확도(accuracy)	99.1%	99.0%
정밀도(precision)	100%	50%
재현율(recall)	10%	90%

• Evaluating model performance is not a simple matter – considering the

model's intended use, Model B is the better model in this case.

• Even in uncertain cases, it is desirable to identify them as aliens to increase recall – depending on the situation, performance should be evaluated by comprehensively considering precision, recall, etc.

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F1 score

- Accuracy becomes a meaningless metric when the imbalance between positives and negatives is severe – the need for a metric that provides a comprehensive value even when data is imbalanced.

- F1 score – harmonic mean of precision and recall

$$\text{F1 점수} = 2 \times \text{재현율} \times \text{정밀도} / (\text{재현율} + \text{정밀도})$$

– The harmonic mean has the characteristic of decreasing when one side reaches an extreme value.

• Accordingly, it provides a more meaningful value than the simple average. • For example, in the case of 100% recall and 0% precision, the F1 score becomes 0.

- Comparison of Alien Detection Glasses A and B – Model A F1 Score = 2*1*0.1 / (1 + 0.1) = 0.182 – Model B F1 Score = 2*0.5*0.9 / (0.5 + 0.9) = 0.643 ÿ Model B shows a higher F1 score

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Performance indicator calculation code

- Output performance metrics for the OX quiz

예제 ex7-1.py

```
from sklearn.metrics import confusion_matrix
from sklearn.metrics import classification_report
y_true = [0,0,0,0,1,1,1,0,0,1]
y_pred = [0,0,0,1,1,1,0,0,1,1]
print(confusion_matrix(y_true, y_pred))
print(classification_report(y_true, y_pred))
```

```
[[4 2]
 [1 3]]
              precision    recall  f

           0       0.80      0.67
           1       0.60      0.75

    accuracy
   macro avg       0.70      0.71
weighted avg       0.72      0.70
```

– Utilizing sklearn's confusion_matrix and classification_report functions – When 1 is treated as positive, the precision is 60%, recall is 75%, and F1 score is 67% – Support indicates the number of occurrences of that value: expressing that there were 4 instances of 1 – The overall accuracy is calculated as 70%

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Performance indicator calculation code

예제 ex7-1.py

```
from sklearn.metrics import confusion_matrix
from sklearn.metrics import classification_report
y_true = [0,0,0,0,1,1,1,0,0,1]
y_pred = [0,0,0,1,1,1,0,0,1,1]
print(confusion_matrix(y_true, y_pred))
print(classification_report(y_true, y_pred))
```

```
[[4 2]
 [1 3]]
              precision    recall  f1-score   support

           0       0.80      0.67      0.73         6
           1       0.60      0.75      0.67         4

    accuracy                           0.70        10
   macro avg       0.70      0.71      0.70        10
weighted avg       0.72      0.70      0.70        10
```

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ROC curve and AUC

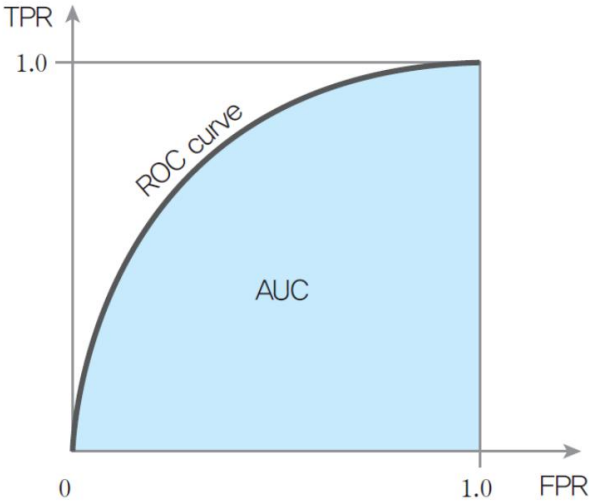
- True Positive Rate (TPR) and False Positive Rate (FPR)
  - True Positive Rate: A synonym for recall rate

$$TPR = \frac{TP}{TP + FN}$$

  - False Positive Rate: The rate of negatives incorrectly identified as positive.

$$FPR = \frac{FP}{FP + TN}$$

- ROC curve (ROC: Receiver Operating Characteristic)
  - Visualization of classification model performance based on classification thresholds
  - Graph showing TPR and FPR according to the threshold at which the model distinguishes between positive and negative



[그림 7-3] ROC 곡선의 예

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ROC example

- Height and weight of patients with metabolic syndrome

  – Severe obesity if BMI (= weight / height²) is 30 or higher –

  What if severe obesity had to be determined based solely on weight?

〈표 7–8〉 ROC 예제 데이터

환자번호	몸무게(kg)	키(cm)	BMI	진단결과
1	58	160	22.7	
2	62	163	23.3	
3	69	151	30.3	고도비만
4	73	175	23.8	
5	77	178	24.3	
6	84	164	31.2	고도비만
7	88	165	32.3	고도비만
8	92	193	24.7	
9	96	172	32.4	고도비만
10	104	183	31.1	고도비만

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ROC Examples (Continued)

• When the weight criterion for determining severe obesity is set at 80kg

〈표 7-9〉 임계값 80kg의 혼동행렬

Actual \ Predicted	0 (정상)	1 (고도비만)
0 (정상)	4 (TN)	1 (FP)
1 (고도비만)	1 (FN)	4 (TP)

– TPR = 4/(1+4) = 0.8, FPR = 1/(1+4) = 0.2

• If the threshold is set to 55kg? ÿ TPR = FPR = 1 • If the threshold is set to 105kg? ÿ TPR = FPR = 0 • If you calculate TPR and FPR while varying the threshold and connect the points, you get an ROC curve

This becomes

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ROC graph code

- Calculate TPR,

  FPR – Import confusion_matrix module from sklearn.metrics

```
from sklearn.metrics import confusion_matrix
```

– Put the patient's weight into a list named 'weight' and height into a list named 'height'

```
weight = [58,62,69,73,77,84,88,92,96,104]
height = [160,163,151,175,178,164,165,193,172,183]
```

– Assign the correct answer value indicating whether each patient is morbidly obese to y_true (0: normal, 1: morbidly obese)

```
y_true = [0,0,1,0,0,1,1,0,1,1]
```

– Set the weight threshold for determining severe obesity to 80 and store it in a variable called threshold.

```
threshold=80
```

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ROC Graph Code (Continued)

- Calculate TPR, FPR

  (continued) – For each patient, assign 1 to y_pred if weight is above the threshold, and 0 otherwise.

```
y_pred=[]
for i in range(10):
    w = weight[i]
    if w>=threshold: y = 1
    else: y = 0
    y_pred.append(y)
```

  – Compare the predicted y_pred with the correct value y_true to output the confusion matrix

```
cm = confusion_matrix(y_true, y_pred)
print(cm)
```

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ROC Graph Code (Continued)

- Calculate TPR, FPR (continued)
  - Read TN, TP, FN, and FP from this confusion matrix

```
tn = cm[0][0]
tp = cm[1][1]
fn = cm[1][0]
fp = cm[0][1]
```

  - Calculate and output TPR and FPR values

```
tpr = tp / (tp+fn)
fpr = fp / (fp+tn)
print(tpr, fpr)
```

```
[[4 1]
 [1 4]]
0.8 0.2
```

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ROC Graph Code (Continued)

- ROC Curve Output Program –

  The part for importing modules and defining data is the same.

```
from sklearn.metrics import confusion_matrix

weight = [58,62,69,73,77,84,88,92,96,104]
height = [160,163,151,175,178,164,165,193,172,183]
y_true = [0,0,1,0,0,1,1,0,1,1]
```

  – Initialize an empty list to store FPR and TPR based on changes in the threshold.

```
fpr_list=[]
tpr_list=[]
```

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ROC Graph Code (Continued)

- ROC Curve Output Program (Continued) –

  Calculate y_pred while varying the threshold value from 55 to 105 in increments of 5, calculate the corresponding FPR and TPR, and store them in fpr_list and tpr_list in order.

  - (Lines 2–14) For a given threshold value

    Calculate the tpr and fpr values

  - (Line 15) Print the values of threshol, tpr, and fpr

  - (Lines 17–18) Write the calculated fpr and tpr to fpr_list and tpr_list, respectively.

```
1  for threshold in range(55,110,5):
2      y_pred=[]
3      for i in range(10):
4          w = weight[i]
5          if w>=threshold: y = 1
6          else: y=0
7          y_pred.append(y)
8      cm = confusion_matrix(y_true, y_pred)
9      tn = cm[0][0]
10     tp = cm[1][1]
11     fn = cm[1][0]
12     fp = cm[0][1]
13     tpr = tp / (tp+fn)
14     fpr = fp / (fp+tn)
15     print(threshold, fpr, tpr)
16
17     fpr_list.append(fpr)
18     tpr_list.append(tpr)
```

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ROC Graph Code (Continued)

- ROC Curve Output Program (Continued) – An ROC

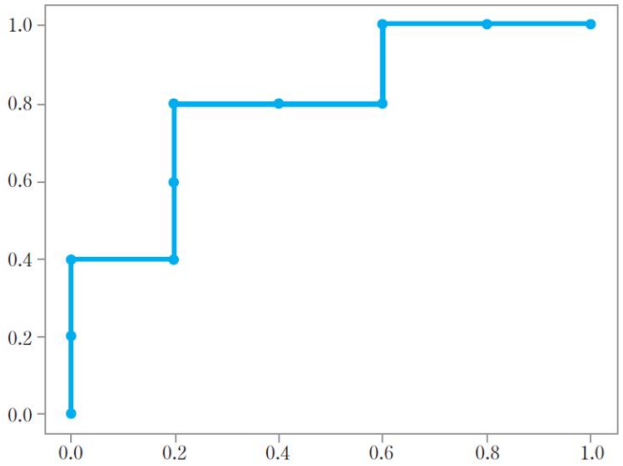
  curve is drawn by plotting and connecting points with the calculated fpr_list on the horizontal axis and tpr_list on the vertical axis.

```
1  import matplotlib.pyplot as plt
2  plt.plot(fpr_list, tpr_list, '-o')
3  plt.show()
```

  - (Line 2) Use the -o option to connect lines and display each point

- Output result

```
55 1.0 1.0
60 0.8 1.0
65 0.6 1.0
70 0.6 0.8
75 0.4 0.8
80 0.2 0.8
85 0.2 0.6
90 0.2 0.4
95 0.0 0.4
100 0.0 0.2
105 0.0 0.0
```



[그림 7-4] 예제프로그램 7-3의 ROC 곡선 출력

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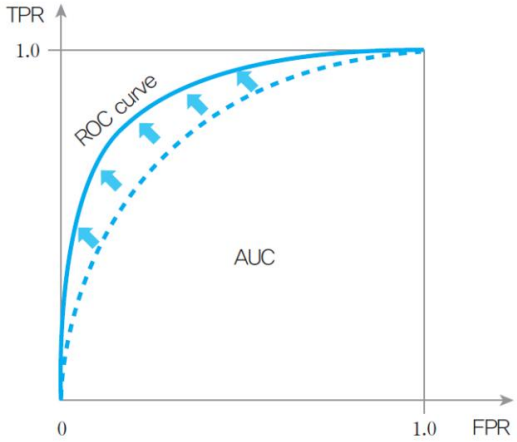
ROC Graph Code (Continued)

- Model Performance and ROC –

  Since the more accurate the prediction model, the higher the TPR and the lower the FPR,
  The better the model, the more the points shift to the upper left.
  The area under the ROC curve, AUC, becomes wider.



[그림 7-5] 모델 개선과 ROC와 AUC의 변화

- Improved model: Determines severe obesity by calculating BMI index

예제 ex7-4.py

```
...
for threshold in np.arange(20, 40, 0.1):
    y_pred=[]
    for i in range(10):
        bmi = weight[i] / (height[i]/100)**2
        if bmi>=threshold: y = 1
        else: y=0
...
```

range can only be incremented by integers, so use np.arange

Instead of body weight Determined as 1 or 0 using the BMI formula

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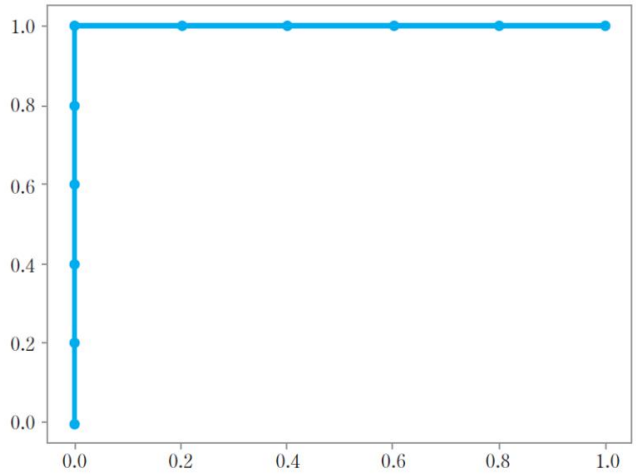
ROC Graph Code (Continued)

- Improved Model
  (Continued) – All points move to the top left and stick until they can no longer move.
  - Since the diagnostic criterion for severe obesity is the BMI index, calculating and predicting the BMI is exactly the same as the correct answer.



[그림 7-6] 개선된 모델의 ROC 곡선

– What is the best threshold value?
  - For the BMI threshold range of 24.7 to 30.2, FPR was 0 and TPR was 1, indicating a perfect judgment without errors. • This implies that the threshold value for determining obesity based on BMI should be between 24.9 and 30.2.
    – In fact, the BMI criterion for severe obesity is approximately 30.0